

SMASH - CAMPAIGN LEVEL DESIGN BLUEPRINT

Mission 1 renovation + Mission 2 (beach) & Mission 3 (space), with image prompts
Companion to SMASH_JSON_Cloning_Handbook.txt (build rules) and the map spec
Project: Demolition Contracts (FierceTigerGameJam) Revision 3 - 2026-08-30

WHO THIS IS FOR

The designer and the image generator producing concepts for Mission 2 and Mission 3, and the AI that will clone those concepts into map JSON. Every level below carries: the archetype, the grid, the material band, the "collapse story" (what makes destroying it feel good), and a ready-to-use IMAGE PROMPT. Concepts must be drawn at CELL RESOLUTION - one drawn brick = one 0.25 m cell of the stated grid - or the JSON cannot honour them.

1. WHY MISSION 1 READS FLAT - THE DIAGNOSIS THIS FIXES

All nine current buildings are the same organism: a hollow rectangular box with window holes. Every map plays the same way - stand in front, sand the facade down, bricks drop at the foot of the wall. No shot is smarter than another, nothing chains, the structure never betrays itself. Demolition is only fun when the building has a STRUCTURE TO BETRAY: a load path the player can read, cut, and watch fail.

Every level below is therefore built around one of five construction styles, and no level is a plain box.

2. THE FIVE CONSTRUCTION STYLES

- S1 PILLAR PUZZLE - a heavy mass carried by 2-4 slim pillars. The player reads "shoot the legs", cuts them, gravity does the rest. The engine already releases everything above a destroyed support.
Shapes: water tower, elevated tank, grandstand, viaduct.
- S2 DEPTH DOMINO - 3-4 tall free-standing slabs placed one behind another, 1-2 layers apart. A knocked wall topples as one plate and slams the next hard enough to activate it: one good shot fells the whole row. Uses a chain mechanic the engine has that NO current map exercises.
- S3 COMPOUND / DIORAMA - two or three small separate structures on one map (main hall + chimney + shed). Every bullet becomes a target CHOICE, and a falling chimney can maul the building beside it.
- S4 CANTILEVER & ARCH - bridges, gates, overhangs propped by one column: shapes that look unstable and itch to be shot. Physics pays it off.
- S5 SHOT ECONOMY - not a shape but a tuning rule that makes S1-S4 matter: set bulletPickLimit to ~60-70% of the shots needed to sand the building down, so passing REQUIRES structural shots.

MATERIAL BANDS decide where the fun comes from (per mission of nine: levels 1-3 brick only, 4-6 brick+glass, 7-9 brick+glass+concrete):

- brick band -> fun = shape and domino (no material contrast, so silhouettes and chains carry everything)
- glass band -> fun = glass rain: continuous panes riding on the load-bearing band below them, released all at once
- concrete band -> fun = anatomy: visible skeleton, player learns to punch the brick infill and then fell the frame

3. MISSION 2 - BEACH / SEASIDE FAMILY

Theme: a seaside strip - huts, piers, lifeguard towers, a lighthouse. The materials do not change art, so the THEME IS CARRIED BY SHAPE and by how the image generator paints the same three blocks: brick reads as sand-brick and timber, glass as sea-view glazing, concrete as stone piers and breakwater. Everything stands on ground or stilts; stilts are the mission's signature (and free S1 gameplay).

LEVEL 1 - BEACH HUT ROW [brick | S3 | grid 14x6, 5 layers]

Three tiny huts side by side, different roof shapes (flat, stepped, peaked), each raised one row on short stilt blocks. ~110 entries.
Collapse story: three separate toy targets; knocking a hut off its stilts slides it into its neighbour. Lesson: choose targets.
IMAGE PROMPT: "isometric voxel beach huts, three small cube-block cabins with different roofs standing on short stilts in a row, sandy brick tones, white background"

LEVEL 2 - LIFEGUARD TOWER [brick | S1 | grid 8x11, 6 layers]

Small cabin with a peaked roof on four tall slim legs, a 1-cell-wide stair ramp leaning against it. ~130 entries.
Collapse story: the mission's first pure leg-cut - two legs down and the cabin belly-flops; the ramp is a bonus domino that falls separately.
IMAGE PROMPT: "isometric voxel lifeguard tower, small hut with peaked roof on four tall thin legs, long ramp leaning to the ground, sandy brick cubes, white background"

LEVEL 3 - THE PIER [brick | S2 | grid 10x7, 11 layers]

A boardwalk running AWAY from the player: 4 pile bents (frames of two posts + crossbeam) one behind another carrying a 1-row deck, a tiny kiosk at the far end. ~160 entries.
Collapse story: the depth-domino level - fell the front bent and the deck pancakes bent by bent all the way to the kiosk.
IMAGE PROMPT: "isometric voxel wooden pier running into the distance, post-and-beam frames carrying a plank deck, small kiosk at the far end, brick-brown cubes, white background"

LEVEL 4 - BEACH BAR [brick+glass | S4 | grid 12x7, 7 layers]

Open-front bar: heavy flat roof carried by the back wall and two front posts, glass wind-screens filling the sides, loose bottle blocks on the counter. ~170 entries.
Collapse story: shoot a front post and the whole roof slaps down onto the counter; the glass screens pop as it comes.
IMAGE PROMPT: "isometric voxel beach bar, open front with flat roof on two posts, glass panels on the sides, counter with small blocks on it, sandy bricks and blue glass cubes, white background"

LEVEL 5 - SEAVIEW RESTAURANT [brick+glass | S4 | grid 12x9, 7 layers]

Two-storey block: solid brick ground floor, ONE FULL GLASS STOREY (panorama dining room), brick roof deck with loose parasol stubs. ~200.
Collapse story: the glass-waist weak seam by the sea - break the dining row and the roof deck drops through it.
IMAGE PROMPT: "isometric voxel seaside restaurant, brick ground floor, entire middle storey of blue glass, rooftop terrace with small blocks, cube construction, white background"

LEVEL 6 - WATERSLIDE TOWER [brick+glass | S3 | grid 14x11, 7 layers]

A stair tower with a glass-tube slide (stepped diagonal run of glass cells) descending to a splash pool rim; separate ticket kiosk. ~220.
Collapse story: the slide is a cantilevered glass ribbon - cut its two support posts and the whole tube rains down the tower's flank.
IMAGE PROMPT: "isometric voxel water slide tower, brick stair tower with a stepped diagonal blue glass tube sliding down to a small pool, ticket kiosk beside it, cube blocks, white background"

LEVEL 7 - LIGHTHOUSE [all three | S1 | grid 9x14, 6 layers]
Concrete rock base, tapering brick shaft (footprint steps in twice), glass lantern room, loose concrete gallery caps. Keeper's cottage attached at the foot. ~230 entries.
Collapse story: the tallest topple in the mission - weaken one side of the shaft and the lantern rides the whole tower down onto the cottage.
IMAGE PROMPT: "isometric voxel lighthouse, grey stone base, tapering brick tower, glass lantern room on top, small cottage at the foot, cube blocks, white background"

LEVEL 8 - SEA FORT [all three | S4 | grid 12x10, 8 layers]
Squat concrete casemate on a stone plinth, brick barracks inside the open court, glass observation slit band, loose battlement caps. ~270.
Collapse story: anatomy - concrete shell resists, so the player learns to shoot through the slit band and gut the brick interior first.
IMAGE PROMPT: "isometric voxel coastal fort, low grey concrete bastion with a horizontal glass slit, brick buildings inside the courtyard, cube blocks, white background"

LEVEL 9 - BOARDWALK RESORT (BOSS) [all three | S3+S1 | grid 16x12, 10 layers]
The strip in one map: lifeguard tower (S1 legs) + beach bar (S4 roof) + a short pier stub (S2 bents) sharing the beach. ~340 entries.
Collapse story: graduation - three structures, three styles, and the pier deck can be dropped onto the bar's roof.
IMAGE PROMPT: "isometric voxel beach resort scene, lifeguard tower on legs, open beach bar, short wooden pier, sandy bricks blue glass grey stone cubes, white background"

4. MISSION 3 - SPACEPORT / ORBITAL FAMILY

Theme: a planetary spaceport - landed ships, domes, gantries, station modules on cradles. HARD RULE: the engine has gravity and no map may float, so every "space" structure is GROUNDED - landed, docked, or on pedestal cradles. That constraint is the mission's gameplay: almost everything stands on legs, and legs are for cutting. Palette carries the theme: concrete grey = hull and armour, glass blue = domes and viewports, brick = painted cargo modules and scaffolds.

LEVEL 1 - CARGO HANGAR [brick | S4 | grid 14x8, 7 layers]
Barrel-profile hangar (stepped arch roof), one huge door opening filling most of the front, stacked loose crate blocks inside. ~140 entries.
Collapse story: the arch IS the building - chip a haunch and the roof ring collapses into the crate stack.
IMAGE PROMPT: "isometric voxel spaceport hangar, stepped arch roof over a huge door opening, stacked cargo crates inside, painted brick cubes, white background"

LEVEL 2 - LAUNCH GANTRY [brick | S2 | grid 10x13, 6 layers]
A tall open scaffold tower (ladder-like frame, many gaps) beside a simple stacked rocket silhouette; two umbilical arms bridge the gap. ~160 entries.
Collapse story: fell the gantry INTO the rocket - the mission opens with its own little disaster movie. Arms are cantilever bonuses.
IMAGE PROMPT: "isometric voxel launch tower, tall open scaffold frame next to a simple stacked rocket, two arms connecting them, brick cubes, white background"

LEVEL 3 - ROCKET ON LEGS [brick | S1 | grid 9x13, 7 layers]
A fat stacked rocket (3-step taper, fins as buttresses) standing on three 2-row legs over a launch ring. ~170 entries.
Collapse story: pure leg-cut with the tallest payoff so far; the taper means the fall sheds loose nose blocks all the way down.
IMAGE PROMPT: "isometric voxel cartoon rocket standing on three legs"

over a launch ring, stepped tapering body with fins, brick cubes, white background"

LEVEL 4 - OBSERVATORY [brick+glass | S4 | grid 11x10, 8 layers]

Drum tower with a stepped glass dome cap and a glass slit down the dome's front; attached low lab annex. ~200 entries.

Collapse story: the dome is a glass avalanche waiting for its drum rim to be broken - one clean shot under the cap empties the whole bowl.

IMAGE PROMPT: "isometric voxel observatory, round tower with a stepped blue glass dome and a vertical slit, small annex building, cube blocks, white background"

LEVEL 5 - BIODOME PAIR [brick+glass | S3 | grid 15x8, 8 layers]

Two stepped glass domes of different sizes on brick ring foundations, linked by a glass tube corridor. ~220 entries.

Collapse story: two shatter targets and a fragile bridge; felling the big dome's ring drags the corridor out of the small one.

IMAGE PROMPT: "isometric voxel greenhouse domes, two stepped blue glass domes on brick ring bases connected by a glass tube corridor, cube blocks, white background"

LEVEL 6 - CONTROL TOWER [brick+glass | S1 | grid 9x13, 7 layers]

Airport-style mushroom tower: slim brick stem, flared deck, wrap-around glass cab on top, loose antenna blocks. ~200 entries.

Collapse story: the definitive stem-cut - the glass cab rides the deck down and detonates on the pad.

IMAGE PROMPT: "isometric voxel control tower, thin brick stem with a wider deck on top wrapped in blue glass, antennas above, cube blocks, white background"

LEVEL 7 - LANDED SHUTTLE [all three | S1 | grid 14x9, 8 layers]

A grey concrete fuselage (stepped nose, stub wings as cantilevers) on four slim landing legs, glass cockpit band, brick cargo pods slung under the wings. ~260 entries.

Collapse story: cut the legs and the ship belly-flops, wings snap at the root, cockpit glass sprays - the mission's poster moment.

IMAGE PROMPT: "isometric voxel landed spaceship, grey blocky fuselage with stub wings on four thin landing legs, blue glass cockpit stripe, brick cargo pods under the wings, cube blocks, white background"

LEVEL 8 - STATION SEGMENT YARD [all three | S4 | grid 15x10, 8 layers]

A station core cylinder resting on two concrete cradle stands, radial brick modules docked to it, two thin arms carrying flat glass solar panels (cantilevered). ~300 entries.

Collapse story: anatomy + cantilever - pop the solar wings for cheap glass rain, then break a cradle and the whole core rolls apart.

IMAGE PROMPT: "isometric voxel space station module in a ground assembly yard, grey cylinder core on two cradle stands, brick side modules, thin arms holding flat blue solar panels, cube blocks, white background"

LEVEL 9 - THE SPACEPORT (BOSS) [all three | everything | grid 16x13, 10 layers]

The whole port: landed shuttle (S1 legs) + control tower (S1 stem) + gantry with rocket (S2 domino), sharing one pad. ~380 entries.

Collapse story: the final exam - three leg-cuts and a domino on one map, and the gantry can be dropped across the shuttle's tail.

IMAGE PROMPT: "isometric voxel spaceport, landed blocky spaceship on legs, mushroom control tower with glass cab, scaffold gantry beside a small rocket, grey blue and brick cubes, white background"

NOTE - previous families: the industrial and urban high-rise families from Revision 2 remain valid designs and can return as Missions 4-5; this revision replaces the themes, not the rules they were built on.

5. MISSION 1 - RENOVATION PLAN (the nine existing maps)

Mission 1 is NOT rebuilt. Its eight buildings already exist, are wall-split, validated and baked; throwing them away wastes work and the mission's job is to teach. The renovation gives each level exactly ONE collapse story and one lesson, mostly through small JSON edits - only level 8 is replaced outright. Mission identity: the village/residential family, "learning the verbs".

LV	map	verdict	renovation
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1	hollow_brick_hut	KEEP	Tutorial box is fine as the one allowed box. Lesson: walls fall.
2	brick_courtyard_tower	RETROFIT	Top row -> loose crenellation caps (D4). Lesson: debris rains.
3	three_storey_brick_house	RETROFIT	Add one interior floor slab per storey (D1). Lesson: pancake.
4	glass_belt_warehouse	KEEP	Already the glass-rain intro (D3); its belt is the lesson.
5	two_storey_glass_house	RETROFIT	Consolidate the scattered windows into ONE continuous glass band at mid-height (D5). Lesson: aim at the weak seam.
6	large_glass_school	RETROFIT	Widen windows to 2x2 panes, loose parapet caps on the roof edge. Lesson: choose targets.
7	concrete_frame_school	KEEP	Already the anatomy intro (A7): frame + infill reads clearly.
8	test_03 (AI legacy file)	REPLACE	The one rebuild in the mission. New: VILLAGE GATEHOUSE - two brick towers, arched gate, glass lantern room over the arch, concrete plinths under the towers. All three materials, S4+S1. Grid 12x10, 7 layers, ~260 entries. IMAGE PROMPT: "isometric voxel gatehouse, two brick towers with a tall arched gate between them, small glass room above the arch, grey concrete bases, cube blocks, white background"
9	hollow_concrete_tower	RETROFIT	Loose chimney pots + antenna blocks on the roof (D4); keep the concrete corners as the first real leg-cut tease.

SHOT ECONOMY PASS (S5) - applies to the whole mission, zero JSON changes: bulletPickLimit is currently generous enough to sand every building down, which is the single biggest source of "phá hơi chán". Retune to ~60-70% of the sanding cost, e.g.: LV1 10, LV2 12, LV3 14, LV4 14, LV5 16, LV6 16, LV7 18, LV8 20, LV9 20 - then adjust from playtests. Passing should require at least one structural shot per map from LV3 on.

Executability note: every RETROFIT above is a small JSON edit on an existing file (the ids, slots and configs do not change; re-bake after). The only image the generator owes for Mission 1 is the LV8 gatehouse.

6. RULES FOR THE IMAGE GENERATOR

- I1 CELL RESOLUTION, ALWAYS. One drawn cube = one 0.25 m cell of the stated grid. If the drawing shows bricks smaller than the grid, the JSON cannot honour it (proven failure - see the render-vs-JSON feedback note in Docs).

- I2 CAMERA: isometric-ish, from front-right, elevated ~20-25 degrees - the same view as the voxel validation renders, so concept and render can be compared side by side.
- I3 PALETTE: brick = warm red-brown, glass = pale sky blue, concrete = neutral grey. Nothing else is buildable.
- I4 BACKGROUND: plain white or flat colour. No environment, no people, no vehicles - the JSON has no cells for them.
- I5 COUNT WHAT YOU DRAW: the drawn footprint and height must fit the stated grid; the stated entry budget is the ceiling.
- I6 ONE IMAGE PER LEVEL, plus (for S2/S1 levels) an optional second image showing the intended collapse direction as arrows.

HANDOFF LOOP (per level): image -> approve -> clone to JSON per the Cloning Handbook -> validate -> voxel render -> compare render to image -> ship JSON + render + manifest. The render, not the image, is the truth.

END